

Efrén López-Morales

P.O. Box 30001
Las Cruces, NM 88003
✉ elopezm@nmsu.edu
🌐 efrénlopez.org

Professional Appointments

2025–Present **Assistant Professor**, *Computer Science*, New Mexico State University, Las Cruces, NM

Education

2021–2025 **Ph.D. in Computer Science**, *Texas A&M University-Corpus Christi*

Advisor: Carlos Rubio-Medrano

Thesis: *CTI-for-CPS: Securing Cyber-Physical Systems via Advanced Cyber Threat Intelligence Methods*

2018–2020 **Master's Degree in Computer Science**, *Arizona State University*

Advisors: Gail-Joon Ahn, Adam Doupé, and Carlos Rubio-Medrano

Thesis: *HoneyPLC: A Next-generation Honeypot for Industrial Control Systems*

2009–2012 **Bachelor's Degree in Computer Networks**, *Universidad Autónoma de Guadalajara*

Publications

Conference Papers with Proceedings (refereed)

Efrén López-Morales*, Ulysse Planta*, Gabriele Marra, Carlos González, Jacob Hopkins, Majid Garoosi, Elias Obreque, Carlos Rubio-Medrano, and Ali Abbasi, "HoneySat: A Network-based Satellite Honeypot Framework," In *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, San Diego, CA, USA, February 2026.

*Co-first authors

Fangzhou Dong, Arvind S Raj, **Efrén López-Morales**, Siyu Liu, Yan Shoshitaishvili, Tiffany Bao, Adam Doupé, Muslum Ozgur Ozmen, and Ruoyu Wang, "Discovering Blind-Trust Vulnerabilities in PLC Binaries via State Machine Recovery," In *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, San Diego, CA, USA, February 2026.

Efrén López-Morales, Ulysse Planta, Carlos E. Rubio-Medrano, Ali Abbasi, Alvaro Cardenas, "SoK: Security of Programmable Logic Controllers," In *Proceedings of the 33rd USENIX Security Symposium*, Philadelphia, PA, USA, August 2024.

Bhupendra Acharya, Dario Lazzaro, **Efrén López-Morales**, Adam Oest, Muhammad Saad, Antonio Emanuele Cinà, Lea Schönherr, Thorsten Holz, "The Imitation Game: Exploring Brand Impersonation Attacks on Social Media Platforms," In *Proceedings of the 33rd USENIX Security Symposium*, Philadelphia, PA, USA, August 2024.

Efrén López-Morales, Carlos E. Rubio-Medrano, Adam Doupé, Yan Shoshitaishvili, Ruoyu Wang, Tiffany Bao, and Gail-Joon Ahn, "HoneyPLC: A Next-generation Honeypot for Industrial Control Systems," In: Bao, T., Tambe, M., Wang, C. (eds) *Cyber Deception. Advances in Information Security*, vol 89. Springer, Cham, October, 2022.

Efrén López-Morales, Carlos E. Rubio-Medrano, Adam Doupé, Yan Shoshitaishvili, Ruoyu Wang, Tiffany Bao, and Gail-Joon Ahn, "HoneyPLC: A Next-generation Honeypot for Industrial Control Systems," In *Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security 2020*, Online, November 2020.

Workshop Papers with Proceedings (refereed)

Efrén López-Morales, "Orbital Escalation: Modeling Satellite Ransomware Attacks Using Game Theory," In *Proceedings of the Workshop on the Security of Space and Satellite Systems (SpaceSec)*, San Diego, CA, USA, February 2026. (32% acceptance rate.)

Efrén López-Morales, Jacob Hopkins, Alvaro Cardenas, Ali Abbasi, Carlos E. Rubio-Medrano, "By the Numbers: Towards Standard Evaluation Metrics for Programmable Logic Controllers' Defenses," In *Proceedings of the 2nd International Workshop on Re-design Industrial Control Systems with Security (RICSS)*, Salt Lake City, UT, USA, October 2024.

Luis Salazar, **Efrén López-Morales**, Juan Lozano, Carlos E. Rubio-Medrano, Alvaro Cardenas, "ICSNet: A Hybrid-Interaction Honeynet for Industrial Control Systems," In *Proceedings of the 6th Joint Workshop on CPS & IoT Security and Privacy (CPSIoTSec)*, Salt Lake City, UT, USA, October 2024.

Posters

Owen Ramsey and **Efrén López-Morales**, "Poster: Mapping the Evidence: A Taxonomy for Satellite Digital Forensics," In *Proceedings of the IEEE Symposium on Security and Privacy (S&P) Posters*, 2026. (39% acceptance rate.)

Honors & Awards

- Feb 2025 **Cyber-Physical Systems (CPS) Rising Star**, National Science Foundation
- Nov 2024 **CAHSI Student Scholar**, Computing Alliance of Hispanic Serving Institutions
- Sep 2023 **Google Conference Scholarship**
- Oct 2021 **CONACYT Doctoral Scholarship**, Mexican National Council for Science and Technology
- Apr 2021 **SAGE Fellowship**, Texas A&M University-Corpus Christi
- 2018 **Fulbright-García Robles Scholarship**, Fulbright COMEXUS

Invited Talks

- Mar 2026 **Preparing Your Thesis Proposal and Becoming a Ph.D. Candidate**, Grad Cohort Workshop Southwest, Computing Research Association (CRA) (Online)
- Apr 2024 **Design and Implementation of a Satellite Honeypot**, 1st Internet of Space Things (IoST) Workshop, University of Chile (Online)
- Mar 2022 **Ciberseguridad: Habilidades y Oportunidades**, Universidad de la Sierra Juárez (Oaxaca, Mexico)

Service

Program Committee

- 2026 ACM Conference on Computer and Communications Security (CCS)
- 2026 IEEE Conference on Resilience and Integrated Security for Space & Critical Systems (RISC)
- 2026 Workshop on Active Defense and Deception (AD&D)

Conference Reviewer

- 2025 ACM Conference on Data and Application Security and Privacy (CODASPY)

Journal Reviewer

- 2026 ACM Transactions on Privacy and Security
- 2025 IEEE Transactions on Industrial Informatics
- 2025 IEEE Security and Privacy
- 2025 IEEE Transactions on Dependable and Secure Computing

Teaching Experience

Instructor

- Spring 2026 CSCI 4205/5205 "Computer Security", *New Mexico State University*
- Fall 2025 CSCI 4240/5240 "Software Reverse Engineering", *New Mexico State University*
- Fall 2023 COSC 4367 "Firewalls and Intrusion Detection Systems", *Texas A&M University-Corpus Christi*

Research Experience

- Summer 2023 **Research Internship**, *CISPA Helmholtz Center for Information Security*
Advisors: Ali Abbasi and Thorsten Holz
- Summer 2022 **Visiting Researcher**, *University of California, Santa Cruz*
Advisor: Álvaro Cárdenas

In the Media

- Feb 27, 2026 **"Hackers Tricked by Fake Satellite in Groundbreaking Cybersecurity Sting"**, featured in *Orbital Today* [🔗](#)
- Mar 27, 2025 **"Security Research Gaps Leave Critical Infrastructure Open to Cyberattack"**, featured in *Communications of the ACM Magazine* [🔗](#)

Industry Experience

- 2020–2021 **Network Support Engineer**, *Cisco Meraki*, Diagnosed and troubleshooted routing and switching, wireless, security, SD-WAN and firewalls

Languages

- Spanish:** Native
- English:** Proficient
- Brazilian Portuguese:** Proficient
- German:** Elementary proficiency